

Compiler Working Group: Report Out

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Terminology

“Compiler IR” – a circuit representation used during compilation (primarily in-memory)

“Interchange Format” – a circuit representation used to connect different systems (primarily serialized)

We will use the term “compile” in preference to the term “transpile”

Key Gaps Identified

FTQC compilation has not been fully demonstrated

Each QPU vendor will need a different circuit format as input

SDKs have different supported circuit formats as output

Goals Identified

Scope will include FTQC/QEC

Scope will include handling hybrid quantum-classical circuits

We expect that algorithms/applications will need to customize compilation

- Use a coarse-grain idea of compilation stages and allow plugins to replace one or more stages or add passes at the end of a stage

Define an OpenQSE preferred compiler IRs

- We expect multiple IRs will be required, especially for FTQC
- Focus on making preferred IRs when they are relevant for interfaces

Define an OpenQSE preferred interchange format

Interaction with Other Working Groups

QRMI/QDMI : Abstract representation of the hardware constraints for compilation & **more TBD**

Would like to follow Qiskit's Lead on distinguishing between **Required**, **Optional** and **Custom**

Required: number of qubits, what is the modality, what interchange formats do you accept

Optional: data about individual qubits

Custom: QPU provider can move whatever data is needed and provide passes that handle them

Milestones (12 months)

Milestone 1 (6 months):

Draft survey paper

Milestone 2 (12 months):

Draft specification of interfaces

Prototype for reference implementation

Deliverables (12 months)

Survey paper

- Understand existing quantum compilers; what IRs they use; what passes they perform
- Define criteria to select an IR
- Include a chart about quantum-classical capabilities
- Is there convergence on IRs? Testing?
- Performance and capabilities in terms of circuit size; how do tools scale with qubit and gate count?

Compiler-related requirements for architecture document

Specification of OpenQSE compiler component

Reference implementation



Working Group Participants

Plan to meet every 2 weeks for ~1 hour each meeting

Expected Participants:

- Michael Ferguson (HPE) (WG lead)
- Narasinga Rao Miniskar (ORNL WG co-lead)
- Matt Treinish (IBM)
- Neal Erickson (Quantinuum)
- Sangram Deshpande (ORNL / student)
- Ulrik De Meulenaere (ORNL / student)
- TBD (Alice & Bob)
- TBD (Amazon Braket)